Questions

Q1.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The Big Bang and Steady State are two theories of the origin of the Universe.

Red shift and cosmic microwave background (CMB) radiation have been discovered by observing the Universe.

Which line of the table links the evidence to the theory it supports?

(1)

		evidence	
	theory	**red shift**	**CMB**
☒ **A**	Big Bang	Yes	No
☒ **B**	Steady State	Yes	Yes
☒ **C**	Big Bang	Yes	Yes
☒ **D**	Steady State	No	Yes

(Total for question = 1 mark)

Q2.

The Big Bang theory and the Steady State theory are two theories about the origin of the universe.

The discovery of CMB led scientists to accept only one of the theories.

Explain why red shift supports both theories but CMB supports only one of them.

(3)

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(Total for question = 3 marks)

Q3.

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

The Steady State theory is also a theory about the origin of the Universe.

Give **one** similarity and **one** difference when comparing the Big Bang theory with the Steady State theory.

(2)

similarity

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difference

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(Total for question = 2 marks)

Q4.

Describe the significance of microwave radiation in the development of a theory of the origin of the universe.

(2)

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(Total for question = 2 marks)

Q5.

Stars may originate as a nebula.

(i) Describe the process that then occurs to produce the conditions necessary for nuclear fusion in a new star.

(3)

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(ii) The energy, E, released in nuclear fusion is equivalent to loss in mass, m, according to the equation.

$$E = mc^2$$

where c is the velocity of light.

$c = 3.00 \times 10^8$ m/s

In 1 second, the energy radiated by the Sun is 3.86×10^{26} J.

Calculate the loss in mass of the Sun in 1 second.

(2)

loss in mass = ... kg

(Total for question = 5 marks)

Q6.

Figure 6 shows two objects, **E** and **D**.

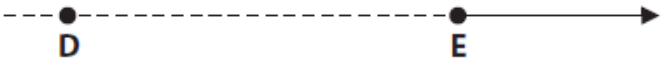

Figure 6

E emits a sound.

D detects the sound.

E emits a sound of wavelength 1.86 m.

D measures the wavelength of this sound as 1.98 m.

(i) Calculate the difference between the wavelength that **E** emits and the wavelength that **D** detects.

(1)

difference in wavelength = ... m

(ii) The velocity of sound is 330 m / s.

Calculate the velocity of **E**.

(2)

Use the equation

$$\text{velocity of } \mathbf{E} = \frac{\text{velocity of sound} \times \text{difference in wavelength}}{\text{wavelength } \mathbf{E} \text{ emits}}$$

velocity of **E** = ... m/s

(Total for question = 3 marks)

Q7.

The table shows some data about the Earth's orbit of the Sun.

orbital radius (R)	1.5×10^8 km
time for one orbit (T)	3.2×10^7 s

Use the following equation to calculate the orbital speed, *v*, of the Earth.

$$v = \frac{2 \times \pi \times R}{T}$$

(3)

v = ... m/s

(Total for question = 3 marks)

Q8.

The Sun has a mass of 2.0×10^{30} kg.
A white dwarf has a mass of 3.4×10^{29} kg.

Calculate the value of

$$\frac{\text{mass of this white dwarf}}{\text{mass of the Sun}}$$

(2)

value = ...

(Total for question = 2 marks)

Q9.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Vesta has an orbital speed of 1.9×10^4 m/s.

Vesta travels a distance of 2.2×10^{12} m when it orbits the Sun once.

Calculate the time taken for Vesta to orbit the Sun once.

(2)

time = ... s

(Total for question = 2 marks)

Q10.

A long time ago scientists thought that the Universe never changed.

Now there is evidence to show that stars progress through various stages and that the Universe is expanding.

(a) Our Sun is in its main sequence stage.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

A star of much greater mass than the Sun will eventually become

(1)

☐ **A** a black hole

☐ **B** a protostar

☐ **C** a red dwarf

☐ **D** a white dwarf

(ii) Describe how the Sun reached its main sequence stage.

(3)

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(iii) Scientists can estimate the age of a star. They want to find the age of the oldest star.

Suggest why knowing the age of the oldest star is not enough to tell scientists the age of the Universe.

(2)

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*(b) Edwin Hubble discovered that the Universe was expanding.
He did this by using observations of red-shift.
Explain what red-shift is and how it provides evidence that the Universe is expanding.

(6)

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(Total for Question = 12 marks)

Q11.

The Sun reaches its highest point in the sky on Midsummer's day.

Describe how you could use this fact to measure the time it takes for the Earth to orbit the Sun once.

(2)

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
	C **both Redshift and CMB can be explained by the Big Bang theory** ***A*** *is not correct because CMB can be explained by the Big Bang* ***B*** *is not correct because the Steady State theory cannot explain CMB* ***D*** *is not correct because the Steady state theory can explain red shift*	**(1)** **AO1**

Q2.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (2 marks): • both theories predict an expanding universe and the Big Bang theory also predicts that the universe had a beginning (1) • the red shift theory indicates that the universe is expanding so supports both theories (1) • whereas CMB also indicates that the universe had a beginning, so supports Big Bang theory (1)	provided evidence that the Steady State theory was incorrect	**(3)**

Q3.

Question number	Answer	Additional guidance	Mark
	similarity (both have) expanding Universe (1) **difference** one from: Steady State, Universe has no beginning (1) Steady State theory requires the continual formation of new matter, the Big Bang theory does not (1)	different interpretation of CMBR	**(2)** **AO1**

Q4.

Question number	Answer	Mark
	An answer that combines 2 points of understanding to provide a logical description: • Discovery of cosmic **background** microwave radiation (1) • Could be explained as resulting from Big Bang (1) • Could not be explained by steady state theory (1)	**(2)**

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	a description to include: nebula collapses (1) under **gravity** (1) plus any one from: GPE converted into KE (1) OR (very) **high** temperatures/pressures reached (1)	 allow gas/dust for nebula allow condensing/coming together for collapses allow gravitational force producing (large) increase in **KE** of particles / more (frequent) collisions Ignore references to hot / heat	**(3)**

Question Number	Answer	Additional guidance	Mark
(ii)	rearrangement **and** substitution (1) (m =) $\frac{3.86 \times 10^{26}}{(3.00 \times 10^{8})^{2}}$ evaluation (1) (m =) 4.29×10^{9} (kg)	 ignore Power Of Ten (POT) error until evaluation allow numbers that round to 4.3×10^{9} (kg) award full marks for the correct answer without working 4.3 to any other power of ten scores 1 mark	**(2)**

Q6.

	Answer	Additional guidance	Mark
(i)	(1.98-1.86) = (+/-) 0.12		**(1)** **AO2**

	Answer	Additional guidance	Mark
(ii)	(velocity =) $\frac{330 \times 0.12}{1.86}$ (1) (+/-) 21.3 (m/s) (1)	ecf from 4ai accept numbers that round to 21 (m/s) award 1,2 marks for (i) and (ii) for the correct answer for (ii) even without working	**(2)** **AO2**

Q7.

Question number	Answer	Additional guidance	Mark
	$v = \frac{2 \times \pi \times R}{T}$ unit conversion (1) 10^8 km = 10^{11} m substitution (1) $v = \frac{2 \times \pi \times 1.5 \times 10^{11}}{3.2 \times 10^7}$ evaluation (1) $v = 2.9 \times 10^4$ (m/s)	Allow values which round to 2.9×10^4 full marks will be awarded for correct numerical answer without working	**(3)**

Q8.

Question number	Answer	Additional guidance	Mark
	substitution (1) $\frac{3.4 (\times 10^{29})}{2.0 (\times 10^{30})}$ evaluation (1) 0.17	award 1 mark for 1.7 to any incorrect power of 10 ignore any units given award full marks for the correct answer without working	**(2)**

Q9.

	Answer	Additional guidance	Mark
	rearrangement and substitution (1) (t =) $\frac{2.2 (\times 10^{12})}{1.9 (\times 10^{4})}$ evaluation (1) 1.2×10^{8} (s)	 allow numbers that round to 1.2×10^{8} e.g. 1.1579×10^{8} award full marks for correct answer without working.	**(2)** **AO2**

Q10.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	A a black hole (1)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	A description including three from: MP1 in a nebula (1) MP2 (particles) attracted / come together by (force of) gravity (1) MP3 pe/ke transferred to thermal/heat energy (gas begins to glow and forms protostar) (1) MP4 until {hot / pressure / dense} enough to start nuclear reaction /fusion (1)	 gas / gas and dust core becomes hot / pressure increases / density increases until fusion of hydrogen starts hydrogen starts to become helium condone "hydrogen burning"	**(3)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	A suggestion involving two from: MP1 the oldest star had not yet appeared when the {Big Bang happened / universe started}(1) MP2 the Universe is older than the oldest star MP3 star takes time to form (1) MP4 can't be certain of this time (1)	stars formed after the Big Bang the age of the oldest star is the minimum age of the Universe estimation is not the same as accurate measurement can't be sure there isn't an older star	**(2)**

Question Number		Indicative Content	Mark
QWC	* **(b)**	An explanation including some of the following points • light shifted to red end of spectrum • light waves are stretched so wavelength increases • reference to black or spectral lines moving to 'red end' (of absorption spectrum) • frequency of wave from a moving source changes • decrease in frequency means source moving away • increase in frequency means source moving towards us • red shift shows galaxies are moving away from us • greater red shift indicates galaxy moving away faster • further away galaxies give greater red shift • (nearly) all galaxies show red-shift • red shift shows decrease in frequency • blue shift shows increase in frequency • therefore galaxies are moving apart • [mention of Doppler effect] • [outline of Doppler effect]	**(6)**

Level	0	No rewardable content
1	1 - 2	• a limited explanation e.g. (light from) {galaxy / planet /object} moving away from us is shifted to red end of the spectrum OR red shift means {galaxy / planet /object) is moving away from us • the answer communicates ideas using simple language and uses limited scientific terminology e.g. correct use of change of colour and movement • spelling, punctuation and grammar are used with limited accuracy
2	3 - 4	• a simple explanation involving detail of meaning of different red shifts OR involving frequency / wavelength e.g. red shift shows galaxies moving away from us. More distant galaxies give greater red shift showing they are travelling faster away. OR light from galaxies/stars moving away is shifted to red end of the spectrum because of an (apparent) {increase in the wavelength/decrease in the frequency} (of light). • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately e.g. correct use of the terms galaxy/star, frequency, wavelength • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	• a detailed explanation correctly interpreting the (apparent) drop in frequency / increase in wavelength e.g. light from (most) galaxies is shifted towards the red end of the spectrum because of an {increase in the wavelength/decrease in the frequency}. This indicates that (most) galaxies are moving away from us, hence showing the Universe is expanding • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately e.g. linkages must be clear between red-shift, movement and expansion of the Universe • spelling, punctuation and grammar are used with few errors

Q11.

Question number	Answer	Additional guidance	Mark
	An answer that combines the following points of application of knowledge and understanding to provide a logical description: • start the clock when the Sun is at the highest point (1) • finish the clock when the Sun is at the highest point again (1)	 Shadow is shortest	**(2)**